

1 · ESXi HOST — physical server, VT-x / AMD-V

2 · LINUX VM — Debian, /dev/kvm exposed to the guest

3 · GITLAB RUNNER — Docker executor, privileged kind-manager image

4 · KIND CLUSTER — Kubernetes nodes as Docker containers

Control-plane node
kube-apiserver · etcd
scheduler · controller-manager

Worker node
kubelet · kube-proxy
containerd

Temporal server
workflow engine
ClusterIP service

Temporal worker
drives the build passes
one per target platform

KubeVirt
virt-operator, virt-controller
virt-launcher pod

Boot disk (PVC)
win.qcow2 · Packer + QEMU
unattended.xml install

Windows VM — running on KVM
full Windows guest on a Linux-only cluster
native C / Rust compilation, no cross-compilation
VirtIO drivers · vCPU / vRAM / vDisk

/dev/kvm passthrough
VT-x / AMD-V
↓ ESXi host
↓ Debian VM
↓ privileged container
↓ kind node
↓ virt-launcher
↓ Windows VM
kernel modules checked to
rule out emulation fallback

Production target
RKE2 on native
Windows and Linux servers

GitLab CI
triggers the job
whole environment rebuilt
from scratch on every run

